Unit II :: Work and Energy

PHY01MX01R04: Mechanics and Properties of Matter

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Syllabus

Unit II: Work and Energy

Concepts of work, power and energy; conservative and non-conservative forces; the work-energy theorem; conservative force as the negative gradient of potential energy; the law of conservation of mechanical energy; work done by non-conservative forces; linear restoring force; potential energy curve and equilibria – stable, unstable and neutral.

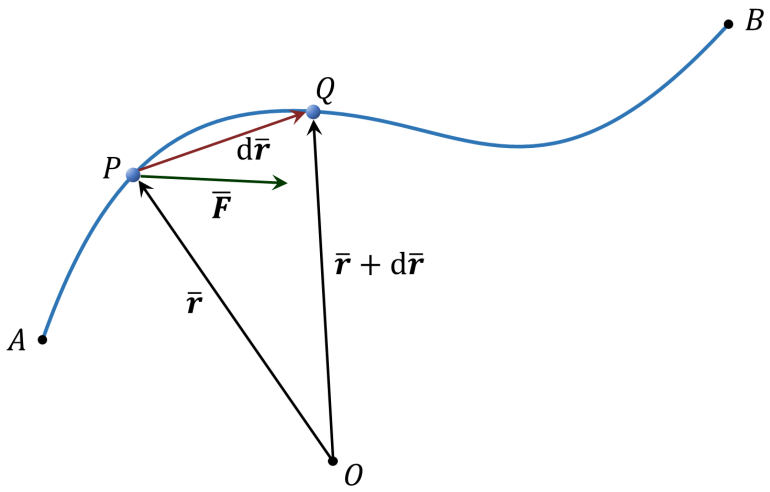
1 Concepts of work, energy and power

Before examining the work-energy theorem and conservation laws in mechanics, we must establish rigorous vector definitions for the foundational quantities of work, energy, and power:

- *Work:* an activity where a force acting on a system causes a displacement.
- *Energy:* the capacity of a system for doing work. A system may possess energy in the form of motion or position.
- *Kinetic energy:* the energy possessed by a system by virtue of its motion.
- *Potential energy:* the energy possessed by a system by virtue of its position or configuration.
- *Power:* the rate at which work is done or energy is used by a system.

1.1 Work

Let us consider a particle of mass m which is moving on an arbitrary path between points A and B. Suppose when the particle is at point P with a position vector $\mathbf{r}$, it is under the influence of some force $\mathbf{F}$ which causes it to get displaced to another point Q with a position vector $\mathbf{r}+\mathrm{d}\mathbf{r}$.



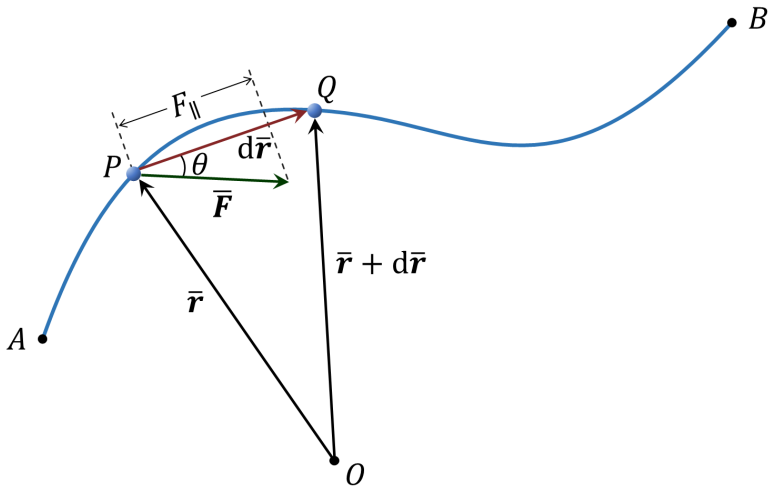
The infinitessimal amount of work done by the force $\mathbf{F}$ in moving the particle from P to Q is

$$\mathrm{d}W = \mathbf{F}\cdot\mathrm{d}\mathbf{r}$$

Hence, the total work done by the force $\mathbf{F}$ in moving the particle from A to B is

$$W = \int_A^B \mathrm{d}W = \int_A^B \mathbf{F}\cdot\mathrm{d}\mathbf{r}$$

1.1.1 Using projection of force vector



If the angle θ between **F** and d**r** is known, then we have

$$dW = F\cos\theta\, dr = F_\parallel\, dr$$

where $F_\parallel = F\cos\theta$ is the projection of **F** along d**r**. Then we can obtain the total work as

$$W = \int_A^B F\cos\theta\, dr = \int_A^B F_\parallel\, dr$$

1.1.2 Using components of force vector

We can write the force vector in component form as follows:

$$\mathbf{F} = \hat{\mathbf{x}}F_x + \hat{\mathbf{y}}F_y + \hat{\mathbf{z}}F_z$$

Also, the infinitessimal displacement is

$$d\mathbf{r} = \hat{\mathbf{x}}\,dx + \hat{\mathbf{y}}\,dy + \hat{\mathbf{z}}\,dz$$

Then the infinitessimal displacement becomes

$$\begin{aligned} dW &= \mathbf{F}\cdot d\mathbf{r} \\ &= \left(\hat{\mathbf{x}}F_x + \hat{\mathbf{y}}F_y + \hat{\mathbf{z}}F_z\right)\cdot\left(\hat{\mathbf{x}}\,dx + \hat{\mathbf{y}}\,dy + \hat{\mathbf{z}}\,dz\right) \\ \implies dW &= F_x\,dx + F_y\,dy + F_z\,dz \end{aligned}$$

Therefore, the total work is obtained as

$$W = \int_A^B \left(F_x\,dx + F_y\,dy + F_z\,dz\right)$$

$$\implies W = \int_A^B F_x\,dx + \int_A^B F_y\,dy + \int_A^B F_z\,dz$$

1.1.3 Units of work

- In the SI system, the unit of work is the **joule** (or J), which is defined as the amount of work done by a force of 1 newton acting through a distance of 1 metre, i.e.

$$1\,\text{J} = 1\,\text{N} \times 1\,\text{m}$$

- In the CGS system, the unit of work is the **erg**, which is defined as the amount of work done by a force of 1 dyne acting through a distance of 1 cm. In other words,

$$1\text{ erg} = 1\text{ dyne} \times 1\,\text{cm}$$

  - The conversion to SI units are as follows:

$$1\text{ erg} = 10^{-7}\,\text{J} \quad \text{and} \quad 1\,\text{J} = 10^{7}\text{ erg}$$

- In the FPS system, the unit of work is the **foot-poundal** (or ft-lbf), which is defined as the amount of work done by a force of 1 poundal acting through a distance of 1 foot or

$$1\text{ ft-lbf} = 1\text{ lbf} \times 1\text{ ft}$$

  - The conversion to SI units are as follows:

$$1\text{ ft-lbf} \approx 1.356\,\text{J} \quad \text{and} \quad 1\,\text{J} \approx 0.736\text{ ft-lbf}$$

1.2 Energy

In classical mechanics, energy is defined as the scalar capacity of a physical system to perform work. Rather than representing a tangible physical substance, energy is a fundamental, conserved state property that quantifies the state of a system relative to external interactions.

The units of energy are exactly the same as that of work.

1.2.1 Kinetic energy

Kinetic energy is the scalar energy possessed by an object due to its state of motion. For a particle of constant mass m moving at instantaneous velocity $\mathbf{v}$ relative to a reference frame, its kinetic energy is expressed as:

$$T = \frac{1}{2} m v^2$$

where $v = |\mathbf{v}| = \sqrt{\mathbf{v}\cdot\mathbf{v}}$ is the instantaneous speed of the particle. Therefore, the kinetic energy of a particle is independent of the direction of motion of the particle.

1.2.2 Potential energy

Potential energy is stored energy which can be attibuted to the relative spatial configuration or position of a particle within a type of force field, known as a *conservative force*. In such a case, the conservative force may be expressed as the negative gradient of the potential energy:

$$\mathbf{F} = -\boldsymbol{\nabla} U$$

Common forms of potential energy include:

- Gravitational potential energy: $U_g = mgh$ (near Earth's surface)
- Elastic potential energy: $U_e = \frac{1}{2}kx^2$ (for an ideal spring satisfying Hooke's Law)

1.3 Power

Power is defined as the rate of doing work. When a force $\mathbf{F}$ displaces a particle by $\mathrm{d}\mathbf{r}$ over a time interval $\mathrm{d}t$, the infinitessimal work done is $\mathrm{d}W = \mathbf{F}\cdot\mathrm{d}\mathbf{r}$. Then the instantaneous power delivered by the force is

$$P = \frac{\mathrm{d}W}{\mathrm{d}t} = \frac{\mathbf{F}\cdot\mathrm{d}\mathbf{r}}{\mathrm{d}t} = \mathbf{F}\cdot\frac{\mathrm{d}\mathbf{r}}{\mathrm{d}t}$$
$$\implies P = \mathbf{F}\cdot\mathbf{v}$$

If some power P is delivered on a particle, the total work done on it is also obtained as

$$W = \int_{t_A}^{t_B} P\,\mathrm{d}t$$

where t_A, t_B are the instants of time when the particle was at positions A and B respectively.

1.3.1 Units of power

- In the SI system, the unit of power is the **watt** (or W), which is defined as the rate at which 1 J of work is done in 1 s, i.e.
$$1\,\mathrm{W} = 1\,\mathrm{J}/1\,\mathrm{s}$$
- In the CGS system, the unit of power is **erg per second** (or erg/s), which is defined as the rate at which 1 erg of work is done in 1 s.
  - The conversion to SI units are as follows:
$$1\ \mathrm{erg/s} = 10^{-7}\,\mathrm{W} \quad \text{and} \quad 1\,\mathrm{W} = 10^{7}\ \mathrm{erg/s}$$
- In the FPS system, the absolute unit of power is **foot-poundal per second** (or ft-lbf/s), which is defined as the rate at which 1 ft-lbf of work is done in 1 s.
  - The conversion to SI units are as follows:
$$1\ \text{ft-lbf/s} \approx 1.356\,\mathrm{W} \quad \text{and} \quad 1\,\mathrm{W} \approx 0.736\ \text{ft-lbf/s}$$
- In practical engineering applications, a commonly used unit of power from the FPS system is the **horsepower** (or hp), defined as
$$1\ \mathrm{hp} = 550\ \text{ft-lbf/s}$$
  - The conversion to SI units are as follows:
$$1\ \mathrm{hp} \approx 0.746\,\mathrm{kW} \quad \text{and} \quad 1\,\mathrm{kW} \approx 1.341\ \mathrm{hp}$$

Exercises

1. Compute the work done in moving a particle from position $(0,1,2)$ to $(5,6,8)$ by the force $\mathbf{F} = \hat{\mathbf{x}}\left(2xy + z^2\right) + \hat{\mathbf{y}}x^2 + \hat{\mathbf{z}}2xz$.

Solution:

The given force is $\mathbf{F} = \hat{\mathbf{x}}\left(2xy + z^2\right) + \hat{\mathbf{y}}x^2 + \hat{\mathbf{z}}2xz$, from which we can identify that

$$F_x = 2xy + z^2,\ F_y = x^2,\ F_z = 2xz$$

Also, identifying the coordinates $(0,1,2)$ and $(5,6,8)$ to be those of the initial and final points A and B respectively, we have the total work done to be

$$\begin{aligned}
W &= \int_A^B F_x\,\mathrm{d}x + \int_A^B F_y\,\mathrm{d}y + \int_A^B F_z\,\mathrm{d}z \\
&= \int_A^B \left(2xy + z^2\right)\mathrm{d}x + \int_A^B x^2\,\mathrm{d}y + \int_A^B 2xz\,\mathrm{d}z \\
&= \int_A^B \left(2xy\,\mathrm{d}x + x^2\,\mathrm{d}y\right) + \int_A^B \left(z^2\,\mathrm{d}x + 2xz\,\mathrm{d}z\right) \\
&= \int_A^B \mathrm{d}\left(x^2y\right) + \int_A^B \mathrm{d}\left(xz^2\right) \\
&= x^2y\Big|_A^B + xz^2\Big|_A^B = \left[x^2y + xz^2\right]_A^B \\
&= x\left[xy + z^2\right]_{x=0,y=1,z=2}^{x=5,y=6,z=8} \\
&= \cdot 5 \cdot \left(5\cdot 6 + 8^2\right) - \cdot 0\left(0\cdot 1 + 2^2\right) \\
&= 5\cdot(30+64) - 0 \\
&= 5\times 94 \\
\therefore\ W &= 470\,\text{units}
\end{aligned}$$

2. A particle is moved from $(-1,0,2)$ to $(3,-8,4)$ by a force $\mathbf{F} = \hat{\mathbf{x}}\left(y^2 - x^2\right) + \hat{\mathbf{y}}2xy$. What is the total work done on the particle?

3. A particle lying in the xy-plane is acted upon by a force directed towards the origin whose magnitude is kr where $r = \sqrt{x^2 + y^2}$ is the distance from the particle to the origin.

   (a) Find the work that must be done to move the particle from the origin to the point $(1,1)$ along the radius vector to that point.

   (b) Find the work that must be done if the particle is first moved to $(1,0)$ and from there to $(1,1)$.

2 Work and energy

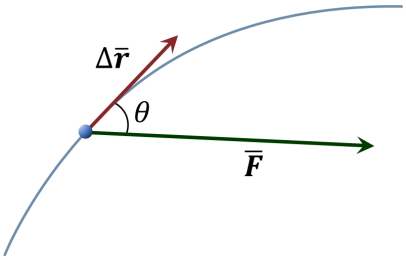
2.1 Work-energy theorem

As per Newton's second law, when a particle of mass m moving with a velocity $\mathbf{v}$ undergoes a force $\mathbf{F}$ at position $\mathbf{r}$, we have

$$\mathbf{F}(\mathbf{r}) = m\frac{\mathrm{d}\mathbf{v}}{\mathrm{d}t}$$

Suppose the particle undergoes a displacement of $\Delta\mathbf{r}$. Then the work done by the force is

$$\Delta W = \mathbf{F}\cdot\Delta\mathbf{r} = m\frac{\mathrm{d}\mathbf{v}}{\mathrm{d}t}\cdot\Delta\mathbf{r} \tag{1}$$



For a time interval Δt which is small enough that $\mathbf{v}$ may be considered to be constant, we can write

$$\Delta\mathbf{r} = \mathbf{v}\Delta t$$

Then equation (1) becomes

$$m\frac{\mathrm{d}\mathbf{v}}{\mathrm{d}t}\cdot\Delta\mathbf{r} = m\frac{\mathrm{d}\mathbf{v}}{\mathrm{d}t}\cdot\mathbf{v}\Delta t \tag{2}$$

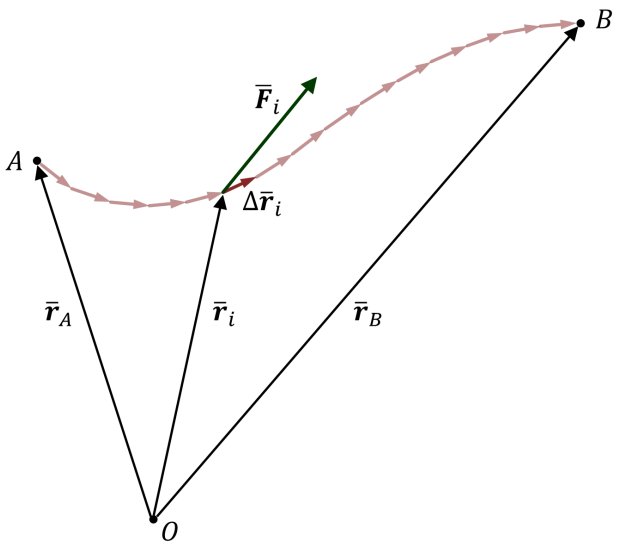
Now, we have

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}t}\left(v^2\right) &= \frac{\mathrm{d}}{\mathrm{d}t}(\mathbf{v}\cdot\mathbf{v})\\ &= \frac{\mathrm{d}\mathbf{v}}{\mathrm{d}t}\cdot\mathbf{v} + \mathbf{v}\cdot\frac{\mathrm{d}\mathbf{v}}{\mathrm{d}t} = 2\frac{\mathrm{d}\mathbf{v}}{\mathrm{d}t}\cdot\mathbf{v}\\ \implies \frac{\mathrm{d}\mathbf{v}}{\mathrm{d}t}\cdot\mathbf{v} &= \frac{1}{2}\frac{\mathrm{d}}{\mathrm{d}t}\left(v^2\right)\end{aligned} \tag{3}$$

Hence, from equations (1), (2) and (3), we have

$$\mathbf{F}\cdot\Delta\mathbf{r} = \frac{1}{2}m\frac{\mathrm{d}}{\mathrm{d}t}\left(v^2\right)\Delta t \tag{4}$$

Suppose now that the path from initial position A to final position B is divided into N small segments. Let $\mathbf{r}_A$ and $\mathbf{r}_B$ be the position vectors of the points A and B respectively.



Now, for the ith displacement at position $\mathbf{r}_i$, if the particle undergoes a force $\mathbf{F}_i$, then the work done over this displacement is obtained from equation (4) to be

$$\Delta W_i = \mathbf{F}_i\cdot\Delta\mathbf{r}_i = \frac{1}{2}m\frac{\mathrm{d}}{\mathrm{d}t}\left(v_i^2\right)\Delta t_i$$

Hence the total work done in moving the particle from A to B is obtained by summing over all the N displacements

$$W = \sum_{i=1}^{N} \Delta W_i = \sum_{i=1}^{N} \frac{1}{2} m \frac{\mathrm{d}}{\mathrm{d}t}\left(v_i^2\right) \Delta t_i \tag{5}$$

As the time intervals $\Delta t_i \to 0$, the summation in equation (5) becomes an integral and we have

$$\begin{aligned} W &= \int_{t_A}^{t_B} \frac{1}{2} m \frac{\mathrm{d}}{\mathrm{d}t}\left(v^2\right) \mathrm{d}t \\ &= \frac{1}{2} m \int_{v_A}^{v_B} \mathrm{d}\left(v^2\right) \\ \implies W &= \frac{1}{2} m v_B^2 - \frac{1}{2} m v_A^2 \end{aligned} \tag{6}$$

In equation (6), we recognise the terms on the RHS to be the kinetic energies of the particle at positions B and A respectively. Hence we can write

$$W = T_B - T_A$$

The above result may be generalised in the form of the *work-energy theorem*, which states that *the total work done on a system equals the change in its kinetic energy*.

$$\boxed{W_{\text{total}} = \Delta T}$$

2.2 Potential energy

3 Conservative and non-conservative forces

3.1 Conservative force as a negative gradient of potential energy

3.2 Law of conservation of mechanical energy

3.3 Work done by non-conservative forces

4 Linear restoring force

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5 Potential energy and equilibria

5.1 Potential energy curve

5.2 Types of equilibria

6 General law of conservation of energy

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